EXPERIMENT – 12
INSTALLATION OF TYPE-2 HYPERVISOR

AIM

To install and study a Type-2 hypervisor (VMware Workstation) on a Windows host machine, and to use it for creating, running and managing a virtual machine.

REQUIREMENTS

1. A computer running the Windows operating system, with hardware virtualization (Intel VT-x or AMD-V) turned on in the BIOS.
2. VMware Workstation installed on the computer.
3. Enough RAM and free disk space to run the virtual machines (at least 8 GB RAM and 30 GB of disk is comfortable).
4. An ISO file of the guest operating system, for example Ubuntu 64-bit.

PROCEDURE

1. Switch on the host computer and log in to Windows. Close the programs that are not needed so that more memory is free.
2. Open Task Manager, go to the Performance tab and select CPU. Check that Virtualization shows as Enabled.
3. Open VMware Workstation using the desktop shortcut or from the Start menu.
4. Look at the Home screen and note the available options: Create a New Virtual Machine, Open a Virtual Machine and Connect to a Remote Server.
5. Select Create a New Virtual Machine to start the wizard (or select Open a Virtual Machine if a VM already exists). Choose the Typical configuration and click Next.
6. Choose “Installer disc image file (ISO)”, browse to the guest operating-system ISO file and click Next.
7. Type a name for the virtual machine and select the folder where its files should be saved.
8. Set the size of the virtual disk (for example 20 GB) and click Next.
9. Click Customize Hardware and set the memory, number of processors and network adapter as required. Then click Close and Finish.
10. Click “Power on this virtual machine” to start it, and follow the installation steps of the guest operating system (language, keyboard layout, disk partitioning and user account).

11. After the installation is over, log in to the guest system and install VMware Tools so that the screen resolution, clipboard and drag-and-drop work properly.
12. Check that the virtual machine is listed in VMware Workstation and that it can be opened, suspended and resumed without any problem.

IMPLEMENTATION INPUT

VMware Workstation application opened on the Windows host computer.

OUTPUT

The VMware Workstation Home screen is displayed successfully with options to create a new virtual machine, open a virtual machine, and connect to a remote server.

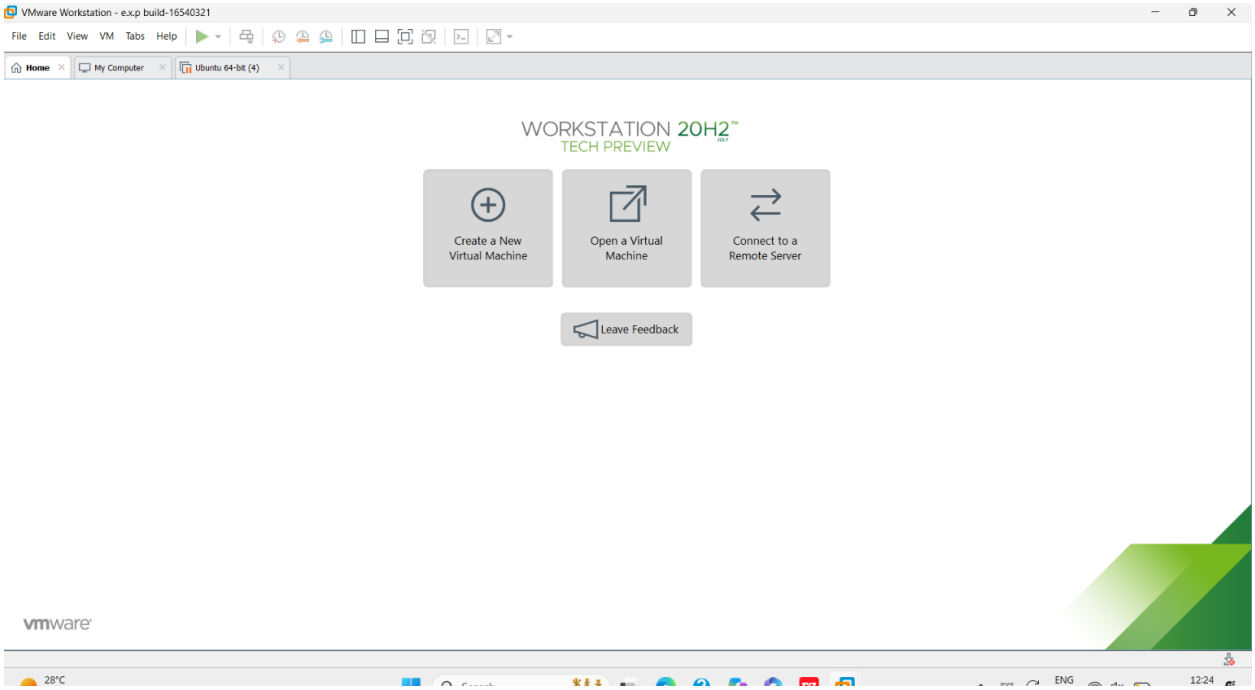


Figure 1: VMware Workstation Home screen (actual screenshot).

RESULT

VMware Workstation, a Type-2 hypervisor, was opened successfully and its virtual-machine management options were verified.